QUALITY ASSURANCE

VWO Login Dashboard

Test Plan

Document version	1.0
Status	Draft for review
Prepared	24 October 2026
Product	VWO authentication and login entry experience (app.vwo.com)
Source	Product Requirements Document supplied by the requester
Test approach	Manual functional and non-functional testing; automation-ready scenarios

Scope note: This plan is based only on the supplied PRD. The PRD combines existing interface observations with proposed capabilities; planned features are tested only when confirmed as implemented and in the release scope. Test against an approved QA or pre-production environment, not production.

Owner, release/build identifier, environment URLs, and execution dates are to be confirmed by the project team.

1. Objective

Verify that the VWO login experience provides secure, understandable, accessible, and reliable entry to the VWO platform. Testing will cover the login form and related recovery and enterprise authentication flows that are included in the release, while checking integration, performance, compatibility, and security expectations stated in the PRD.

The plan is intended to guide test design, execution, defect triage, release-readiness assessment, and test closure. It does not claim that every proposed capability is currently available.

2. Scope

In scope

- Email-and-password sign-in, required-field handling, email-format checks, blur validation, loading feedback, and authentication error recovery.
- Remember Me/session persistence, session expiration and safe return to the intended destination, where supported and specified for the release.

- Password reset and recovery, including the request, reset link/token, password policy, and completion flow when implemented.
- Optional multi-factor authentication (MFA), enterprise SSO (including SAML/OAuth), and social identity providers only when enabled in the approved test environment.
- Registration/free-trial navigation, login announcement/banner, and light/dark theme options described in the PRD.
- Successful transition to the VWO dashboard, session and analytics integration, and login assistance/support links.
- Responsive behavior, keyboard and screen-reader support, high contrast, WCAG 2.1 AA checks, and supported browser/device compatibility.
- Page-load and authentication performance, availability-related checks, HTTPS/session protections, rate limiting, and privacy/compliance-related user-data handling.

Out of scope

- Testing VWO experimentation, personalization, reporting, or analytics features beyond verifying that authentication reaches the correct dashboard and that agreed login events are recorded.
- Account provisioning, billing, trial conversion, or the full post-registration onboarding journey beyond verifying the navigation handoff.
- Biometrics, adaptive authentication, and progressive web app capabilities listed as future enhancements, unless separately added to the release scope.
- Unapproved destructive security testing, production load tests, or assessment of third-party identity providers outside the agreed test tenant.

Release scoping

Phase 1 capabilities (core login and password reset) are the baseline. Phase 2 UX/accessibility and Phase 3 enterprise/security capabilities are included only when scheduled for the release. The product owner must confirm which PRD statements are acceptance requirements before execution.

3. Inclusions and Test Scenarios

The following high-level scenarios provide the initial coverage baseline. Detailed test cases, account data, and expected messages should be reviewed against approved product behavior before execution.

ID	Scenario and expected result	Priority
AUTH-01	Submit valid active-user credentials; authentication succeeds and the user reaches the correct VWO dashboard or approved return URL.	P0
AUTH-02	Submit an incorrect password or unknown account; access is denied, feedback is clear, and response does not unnecessarily disclose account existence.	P0
AUTH-03	Submit empty email/password and whitespace-only values; required-field feedback appears and no unintended authentication request is made.	P1
AUTH-04	Enter malformed email values and valid boundary formats; validation matches the agreed email rules and provides actionable feedback on blur and submission.	P1

AUTH-05	Use keyboard-only navigation, labels, and submit controls; focus order and visible focus are logical and all login actions remain operable.	P1
AUTH-06	Submit the form once and under slow response; a clear pending state is shown, duplicate submission is prevented or safely handled, and outcome is communicated.	P1
AUTH-07	Enable Remember Me and sign in; persistence follows the approved duration and browser/device policy. Disable it and verify the shorter/non-persistent behavior.	P1
AUTH-08	Allow a session to expire or sign out; protected destinations require reauthentication and no stale session grants access.	P0
AUTH-09	Request a password reset for eligible and ineligible addresses; response follows privacy-safe messaging and recovery delivery behavior.	P1
AUTH-10	Use valid, expired, already-used, and malformed reset links; only a valid unexpired token can complete the reset, and token reuse is rejected.	P1
AUTH-11	Set a new password at policy boundaries; accepted/rejected values match approved password requirements, with clear validation and successful subsequent sign-in.	P1
AUTH-12	Complete MFA challenge when enabled; valid, invalid, expired, and retry cases follow configured policy and do not bypass the second factor.	P1*
AUTH-13	Complete SSO with a configured test identity provider; successful, cancelled, failed, and misconfigured responses return safe actionable outcomes.	P1*
AUTH-14	Use each enabled social login provider; successful and denied/cancelled authorization outcomes are handled without creating unintended sessions.	P2*
UX-01	Follow free-trial/registration link and support link; destination is correct, available, and does not lose context unexpectedly.	P2
UX-02	Render announcement/banner and change light/dark mode; controls are visible, readable, persistent as specified, and do not obstruct sign-in.	P2
UX-03	Load login page on desktop, tablet, and mobile viewports; form remains usable without clipped content or unintended horizontal scrolling.	P1

UX-04	Use screen reader, browser zoom, and high-contrast settings; names, errors, status changes, and controls are perceivable and operable.	P1
INT-01	After login, verify dashboard routing and preservation of an approved deep link; unauthorized routes remain protected.	P0
INT-02	Verify agreed login success/failure analytics events; events are accurate, deduplicated as defined, and contain no plaintext password or unnecessary personal data.	P1
SEC-01	Inspect approved authentication traffic and browser storage; credentials/tokens are not exposed in URLs, insecure transport, or unintended client-visible locations.	P0
SEC-02	Verify session cookie/token protections and rotation/logout behavior against the approved security design; replaying an invalidated session does not restore access.	P0
SEC-03	Exercise configured rate limiting using approved test accounts and limits; throttling and recovery behavior work without locking out unrelated users.	P1
SEC-04	Verify authentication failures, reset flows, and audit events do not expose secrets or sensitive personal data in UI responses or logs accessible to testers.	P1
PERF-01	Measure login page load on agreed standard network/device; target is under 2 seconds, with measurement method and percentile agreed before sign-off.	P1
PERF-02	Run approved concurrent sign-in workload in a non-production environment; monitor response time, error rate, and stability against the agreed capacity profile.	P1
COMP-01	Repeat critical login/recovery journeys on the agreed Chrome, Edge, Firefox, Safari, Android Chrome, and iOS Safari versions.	P1

P0 = release-blocking core/security journey; P1 = high priority; P2 = normal priority. Asterisk (*) means test only if the capability is enabled for this release.

4. Test Environments

Environment / matrix	Plan	Details to confirm
QA	Primary functional, integration, accessibility, and security verification.	URL, build/version, test data reset, service dependencies, mail sink, and account setup.

Pre-production	Release-candidate smoke, cross-browser, and approved performance checks.	URL, production parity, identity-provider test tenant, monitoring access, and load-test permission.
Browsers/devices	Current supported Chrome, Edge, Firefox, Safari; Android Chrome and iOS Safari; desktop, tablet, and mobile sizes.	Exact versions, OS versions, viewport list, and support policy.
Network/accessibility	Standard broadband and throttled mobile profile; keyboard, screen reader, browser zoom, and high-contrast mode.	Approved tools, test network profile, assistive technology versions.
Accounts/integrations	Dedicated synthetic accounts for active, disabled, MFA, SSO, and recovery cases; non-production email delivery and identity-provider tenants.	Role matrix, lockout/reset policy, seeded account ownership, and safe cleanup process.

All test accounts and data must be synthetic or explicitly approved. Do not use real customer credentials or personal data. Keep environment configuration and secrets in the approved secret manager.

5. Defect Reporting Procedure

1. Log defects in the team-approved tracker (tool to be confirmed) with title, environment/build, browser/device, account class (never password), preconditions, numbered reproduction steps, expected/actual result, severity, priority, and evidence with secrets/personal data removed.
2. Assign severity: Critical (unauthorized access, broad outage, or unrecoverable core login); High (major authentication/recovery/security failure with no reasonable workaround); Medium (limited functional/accessibility issue or workaround available); Low (minor visual/copy issue).
3. QA triages with product and engineering, links the defect to requirement/test case, and records owner and target fix build. P0/P1 defects are reviewed before release decisions.
4. After a fix, retest the failed case and run the relevant regression set. Reopen if the defect persists; close only with evidence and build recorded.
5. Provide agreed daily/iteration status: executed/pass/fail/blocked counts, open defects by severity, risk, and next actions.

6. Test Strategy

- **Requirement analysis:** map each confirmed PRD requirement to test scenarios; clarify acceptance criteria and unsupported behavior before execution.
- **Test design:** use equivalence partitioning, boundary-value analysis, decision tables for authentication outcomes, state-transition testing for sessions/MFA/reset tokens, use-case testing for end-to-end journeys, and error guessing/exploratory testing.
- **Test levels:** component/API checks by engineering where available; UI functional, integration, end-to-end, compatibility, accessibility, security verification, and non-functional checks by QA.
- **Execution order:** build verification and smoke tests, P0 core flows, planned functional/regression suite, browser/accessibility matrix, integration and security checks, then approved performance run.
- **Automation:** prioritize stable, repeatable P0/P1 journeys and regression coverage in the team's approved Playwright/TypeScript setup. Keep MFA/SSO provider dependencies isolated using test tenants and approved test hooks; do not automate around production controls.

- **Regression:** rerun affected cases after each fix and the full critical login/recovery/session suite for release candidates. Exploratory testing supplements, but does not replace, traceable tests.

7. Test Schedule

Plan work by release/sprint milestones; calendar dates and duration remain TBD until the delivery schedule, build availability, and scope are confirmed.

Activity	Timing / dependency	Exit output
Requirement review and risk assessment	Before test design; product answers open questions.	Approved scope, traceability, environment/data needs.
Test case design and review	During development; complete before feature test start.	Reviewed scenarios and test data plan.
Build smoke and core functional testing	On each QA build; P0 flows first.	Build acceptance result and defect list.
Integration, UX, accessibility, compatibility, security	After stable feature build; repeat on release candidate as needed.	Execution evidence and prioritized defects.
Performance and resilience checks	After capacity profile and non-production authorization are agreed.	Results against approved thresholds.
Regression, closure, and sign-off	Release candidate after fixes.	Test summary, residual risks, approvals.

8. Test Deliverables

- Reviewed test plan and requirement-to-test traceability.
- Prioritized test scenarios/cases, test data and environment matrix.
- Smoke, functional, integration, compatibility, accessibility, security, and performance execution records as applicable to release scope.
- Defect records with sanitized evidence, retest results, and regression status.
- Periodic test status and final test summary including pass/fail/blocked counts, coverage, open defects, known limitations, and release recommendation.

9. Entry and Exit Criteria

Test design entry

- PRD/release scope and acceptance criteria are available; product owner confirms which proposed features ship.

- QA and engineering contacts, risk owner, and defect tracker are identified.

Test design exit

- Requirements and open questions are reviewed; critical paths and risks have traceable scenarios.
- Test data, environment, browser matrix, and success thresholds are approved or explicitly recorded as blockers.

Execution entry

- Deployable build is available in QA with version recorded; dependencies and test accounts are accessible.
- Smoke-test path is available; observability and defect reporting are operational.
- Performance/security testing has written authorization and an agreed non-production profile.

Execution exit / release recommendation

- All in-scope P0 tests pass; all planned tests are executed or have documented owner-approved waiver/blocker.
- No open Critical or High defects remain unless explicitly accepted in writing by the authorized release owner with mitigation.
- Results, traceability, performance/accessibility findings, known issues, and residual risks are reported.
- Proposed quality gate: at least 95% of planned in-scope tests pass, with the remainder documented and approved. Product/release owners must confirm this threshold before execution.

Test closure

- Final execution and defect reports are complete; evidence is stored in the approved project location.
- Release owner and QA lead review the summary and record go/no-go or conditional acceptance.

10. Tools

Need	Tooling
Defect management	Project-approved tracker (Jira is a candidate; confirmation required).
Browser and device coverage	Supported desktop browsers and physical/emulated mobile devices approved by the team.
Automation	Playwright with TypeScript, consistent with the supplied test-plan skill; repository/framework availability to confirm.
Accessibility	Keyboard and screen-reader checks plus an approved automated accessibility scanner; manual verification remains required.
Performance/monitoring	Approved load-test and browser timing tools, test-environment monitoring, and agreed dashboard/reporting.

Documentation/evidence	Approved document repository, spreadsheet/test management, and screenshot/recording tools with sensitive data redacted.

11. Risks and Mitigations

Risk	Impact	Mitigation
PRD mixes observed behavior and future/proposed requirements.	Mis-scoped tests and false failures.	Product owner confirms release scope, acceptance criteria, and feature flags before test design sign-off.
QA/pre-production environment or identity-provider/email integration is unavailable.	Blocked SSO, MFA, and recovery coverage.	Provision isolated test tenants, mail sink, synthetic accounts, access, and fallback schedule before execution.
Password/session, lockout, timeout, and rate-limit policies are unspecified.	Expected behavior cannot be objectively assessed.	Obtain approved policy values and security design; record unresolved criteria as blockers.
Concurrent load testing could affect shared services.	Service degradation or misleading results.	Run only with written authorization in isolated non-production at an agreed ramp/profile; monitor and stop on agreed guardrails.
Browser/OS versions and 2-second page-load measurement are undefined.	Inconsistent compatibility/performance sign-off.	Agree support matrix, network/device baseline, measurement start/end, sample size, and percentile before execution.
Test evidence may contain credentials or personal information.	Privacy/security exposure.	Use synthetic identities, redact evidence, restrict access, and follow retention/deletion policy.
Late changes to authentication or external provider configuration.	Regression and schedule risk.	Use change-impact assessment, repeat affected P0/P1 tests, and keep provider configuration/version in the test record.

12. Approvals

Role	Name	Approval / date

Product owner	TBD	Pending
QA lead	TBD	Pending
Engineering / security representative	TBD	Pending
Release owner	TBD	Pending

Test execution proceeds against the approved scope and environment. Scope changes, waivers, and residual release risks require documented approval by the appropriate owner.

13. Assumptions and Open Questions

1. Which capabilities are in this release: password reset, MFA, SSO (SAML/OAuth), social login, analytics events, support integration, and light/dark mode?
2. What are the approved QA and pre-production URLs, release/build identifier, and test-account roles?
3. What are the password policy, account lockout/retry, rate-limit, session timeout, Remember Me duration, and logout/session revocation rules?
4. What authentication error text and account-enumeration policy are approved? What is the required recovery email delivery and token-expiry behavior?
5. What are the supported browser/OS versions, locales, and assistive technologies?
6. How is the 2-second load target measured (network, device, start/end event, percentile)? What concurrency profile and response/error thresholds define acceptance?
7. What analytics events and data-retention/privacy requirements apply, and who approves GDPR/CCPA evidence?
8. Confirm the proposed 95% test-pass quality gate, severity definitions, defect tracker, schedule, and sign-off authorities.

End of document